EXPERIMENTAL ANALOG CONTROL — one LTC3779 research phase
No activeOV, no milliampere protection, no4-phase sharing. Native power design unchanged.

J1
AUX5 1
PGND 2
FLOATING +5V AUX / PGND

J2
VCMD 1
ISET 2
PGND 3
VCMD / ISET / PGND

J3
VFB 1
ITH 2
ENABLE 3
PGND 4
VFB / ITH / ENABLE / PGND

#FLG1
PWR_FLAG
AUX5

#FLG2
PWR_FLAG
PGND

J4
KS_HIGH 1
KS_LOW 2
PGND 3
PGND
SHUNT KELVIN + / − / PGND

U1A
CV_POS 3
CV_NEG 2
1 CV_DRIVE
OPA4192IPW

U1B
ISET 5
CC_NEG 6
7 CC_OUT
OPA4192IPW

U1C
IREF_RAW 10
IREF 9
8 IREF
OPA4192IPW

U1D
DIFF_POS 12
DIFF_NEG 13
14 VDIFF
OPA4192IPW

U1E
AUX5
4 V+
V− 11
PGND
OPA4192IPW

U2
INP 8
INM 1
IREF 3
IREF 7
IN+ VS 6 AUX5
IN− OUT 5 IMON
REF2 NC 4 PGND
REF1 GND 2 PGND
INA240A2D

U3
PGND 2
PGND 3
A K 1 REF25
NC_/_A
LM4040AIM3-2.5/NOPB

D1
INJ_NODE CV_DRIVE
DFLS1200-7 — VF REVIEW PENDING

D2
CC_OUT ITH_LIM
DFLS1200-7 — VF REVIEW PENDING

VCMD 0..1.2 V -> high-to-low Vout. CV diode prevents intentional feedback sinking.
ISET=0.1+0.05*I; lower than-0.1106 V causes hardware shutdown; PI values unvalidated.

VFB R1 1M 0.1% CV_POS
VCMD R2 1M 0.1% CV_POS
INJ_NODE R3 1M 0.1% CV_NEG
CV_NEG R4 1M 0.1% PGND
REF25 R5 100k VCMD
ISET R6 100k PGND

INJ_NODE R7 10k 0.1% VFB
IMON R8 20k CC_NEG
CC_OUT R9 10k CC_COMP
ITH R10 470 ITH_LIM
REF25 R11 240k 0.1% IREF_RAW
IREF_RAW R12 10k 0.1% PGND

AUX5 R13 2.49k 1% REF25
REF25 R14 10k 0.1% VMID
VMID R15 10k 0.1% PGND
INJ_NODE R16 1M 0.1% DIFF_POS
VMID R17 1M 0.1% DIFF_POS
PGND R18 1M 0.1% DIFF_POS

VFB R19 1M 0.1% DIFF_NEG
VCMD R20 1M 0.1% DIFF_NEG
VDIFF R21 1M 0.1% DIFF_NEG
KS_HIGH R22 0 INP
KS_LOW R23 0 INM

AUX5 C1 100nF 50V X7R PGND
AUX5 C2 100nF 50V X7R PGND
REF25 C7 100nF 50V X7R PGND
CC_COMP C8 1nF 50V C0G CC_NEG

TP1 AUX5
TP2 PGND
TP3 REF25
TP4 VCMD
TP5 INJ_NODE
TP6 VFB
TP7 ISET
TP8 IREF
TP9 IMON
TP10 CC_OUT
TP11 ITH
TP12 ENABLE
TP13 VDIFF
TP14 VMON
TP15 REF_UV
TP16 REF_OV

Fault guard and output monitor
File: faults.kicad_sch

PCB status: routing report; no hardware tests; not for fabrication
1.2..~100 V / 0..10 A commands; ~0.21 A active floor
Source-only VFB injection; sink-only ITH clamp

Sheet: /
File: control.kicad_sch
Title: EXPERIMENTAL CV/CC — SINGLE PHASE ONLY — NOT4kW
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.2 | Id: 1/2

Hardware guards:5V/ref validity, VCMD andISET range, actualINJ andmismatch, VOUT OVP.
Faults may automatically restart when cleared. RUN−off does not discharge/disconnect Cout.

JP1

AUX5 1
ARM_FEED 2

ARM − DEFAULT OPEN

J5

VOUT_SENSE 1
PGND 2

VOUT SENSE / PGND − HV

U5 TPS3808G50DBVR
SENSE
MR
CT
VDD
GND
RESET
AUX5 AUX5 DELAY AUX5 ENABLE PGND

U6 TPS37000DDCR
INA−OUTA
INB−OUTB
REF_UV REF_OV AUX5 ENABLE ENABLE PGND

U6A TLV1704IPW
REF_CMD_HI VCMD ENABLE

U6B TLV1704IPW
REF25 INJ_NODE ENABLE

U6C TLV1704IPW
ISET REF_I_LO ENABLE

U6D TLV1704IPW
REF_I_HI ISET ENABLE

U6E TLV1704IPW
AUX5 PGND

U7A TLV1704IPW
IMON REF_IMON_LO ENABLE

U7B TLV1704IPW
VDIFF REF_DIFF_LO ENABLE

U7C TLV1704IPW
REF_DIFF_HI VDIFF ENABLE

U7D TLV1704IPW
REF25 VMON ENABLE

U7E TLV1704IPW
AUX5 PGND

R24 10k
ARM_FEED ENABLE

R25 10k
ENABLE PGND

R26 10.3k 0.1%
REF25 REF_CMD_HI

R27 10k 0.1%
REF_CMD_HI PGND

R28 216k 0.1%
REF25 REF_I_LO

R29 10k 0.1%
REF_I_LO PGND

R30 30k 0.1%
REF25 REF_I_HI

R31 10k 0.1%
REF_I_HI PGND

R32 324k 0.1%
REF25 REF_IMON_LO

R33 10k 0.1%
REF_IMON_LO PGND

R34 11.8k 0.1%
REF25 REF_DIFF_LO

R35 10k 0.1%
REF_DIFF_LO PGND

R36 8.45k 0.1%
REF25 REF_DIFF_HI

R37 10k 0.1%
REF_DIFF_HI PGND

R38 49.9k 0.1%
REF25 REF_UV

R39 10k 0.1%
REF_UV PGND

R40 56k 0.1%
REF25 REF_OV

R41 10k 0.1%
REF_OV PGND

R42 130k 0.1%
VOUT_SENSE HV1

R43 130k 0.1%
HV1 HV2

R44 137k 0.1%
HV2 VMON

R45 10k 0.1%
VMON PGND

C3 100nF 50V X7R
AUX5 PGND

C4 100nF 50V X7R
AUX5 PGND

C5 100nF 50V X7R
AUX5 PGND

C6 100nF 50V X7R
AUX5 PGND

C9 10nF 50V X7R
DELAY PGND

C10 100pF 50V C0G
VMON PGND

OVP ~101.75 V nominal; no hard 100 V limit
ENABLE default low; manual ARM is the pull−up source
Open−collector fault outputs; no latch/output switch

Sheet: /Fault guard and output monitor/
File: faults.kicad_sch

Title: EXPERIMENTAL FAULT GUARDS

Size: A3	Date:	Rev:
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